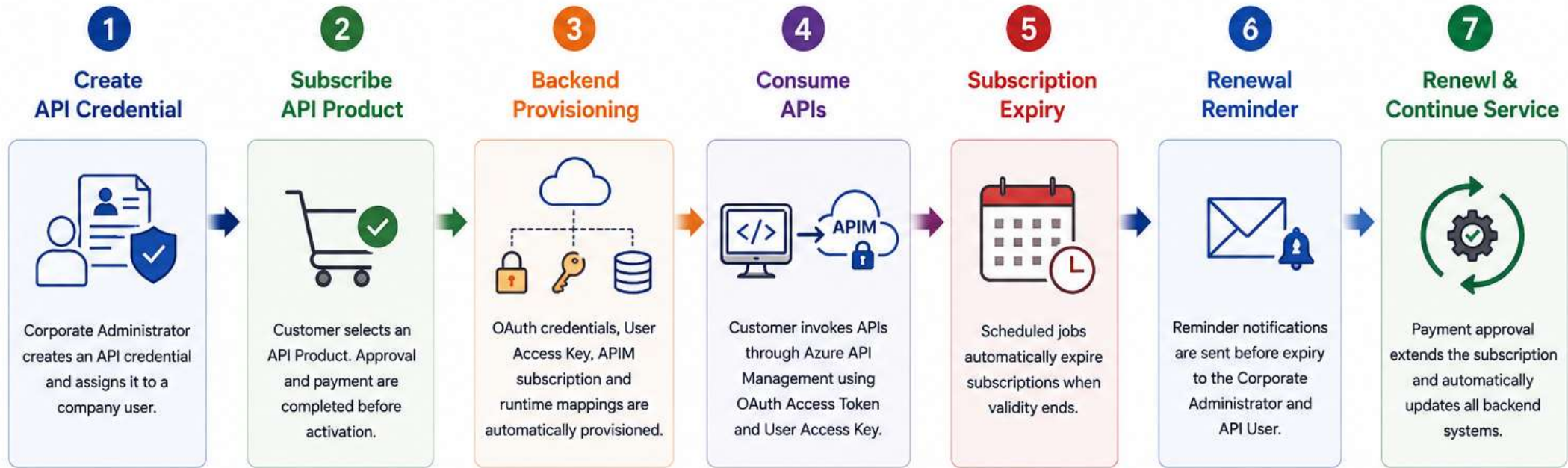


APIM API Subscription Lifecycle Overview

High-Level End-to-End Customer Journey



Lifecycle Summary



Create API Credential

Assign to user with approval



Subscribe API Product

Approval & payment before activation



Backend Provisioning

Automated setup of all backend services



Consume APIs

Secure access via Azure API Management



Subscription Expiry

Automatic expiry via scheduled jobs



Renewal Reminder

Notifications before subscription expiry



Renew & Continue Service

Extend validity and continue API usage



Provision Credential



Subscribe Product



Provision Backend



Use APIs



Expire



Renew



Continue Service

API Credential Provisioning & User Ownership

How existing corporate customers securely create and assign API credentials

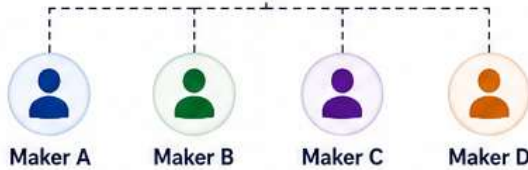
EXISTING CUSTOMER STRUCTURE

Existing Company (Enterprise Customer)

(Enterprise Customer)



Corporate Administrator
(Manages Company Users)



API credentials are owned by the company.

- Existing company with multiple users
- Users managed by the Corporate Administrator
- Each credential is assigned to one user
- Every API call can be traced back to a responsible individual

API CREDENTIAL REQUEST WORKFLOW



Corporate Administrator
Create API Credential

Select



API Product

- Free
- Basic
- Premium



Source IP
Address

Company User
(Maker)



Submit Request



CFO Approval



Back Office Approval



Generate Payment Request

FREE PRODUCT



Auto Approved



Subscription Activated
(Validity Starts)



Trigger Provisioning

BASIC / PREMIUM



Pending CFO Approval



Payment Successful



Subscription Activated
(Validity Starts)



Trigger Provisioning



Maximum 10 API Credentials per company.

CREDENTIAL ASSIGNMENT

Provisioned API Credential



Client ID



Client Secret

Assigned To



John Tan
(Company Maker)



Corporate Administrator
Receives (via Email)

- Client ID
- Client Secret
- Selected API Product
(Free / Basic / Premium)
- Subscription Validity Period
- Development Portal Access



Company Maker
Retrieves from CFO

- Existing User Access Key
(retrieved from the CFO)



User Access Key is
not sent by email.



KEY TAKEAWAY



One Company



Many Users



Request
API Credential



CFO Approval



Back Office
Approval



Payment
(if required)



Subscription Activated
(Validity Starts)



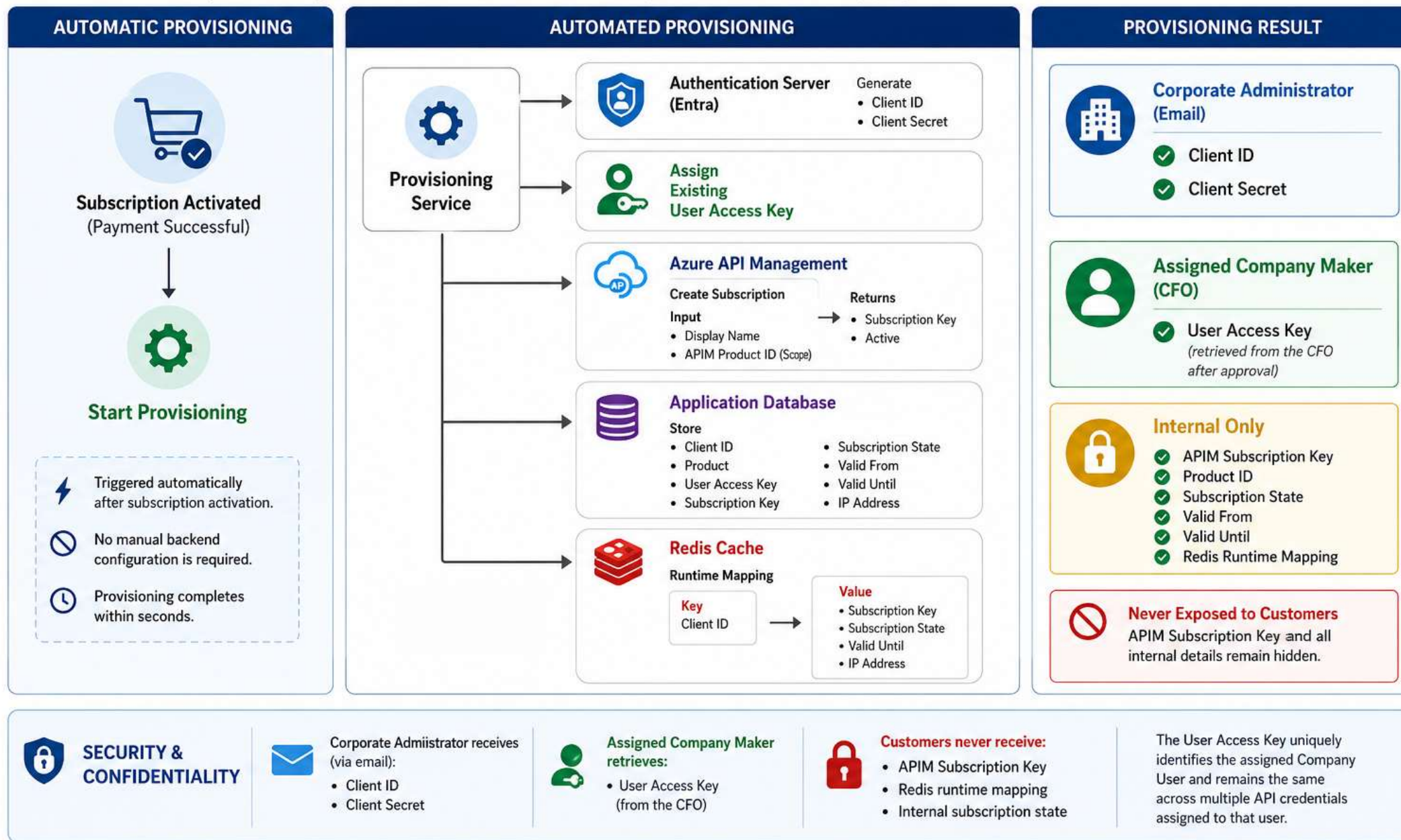
Automatic
Provisioning

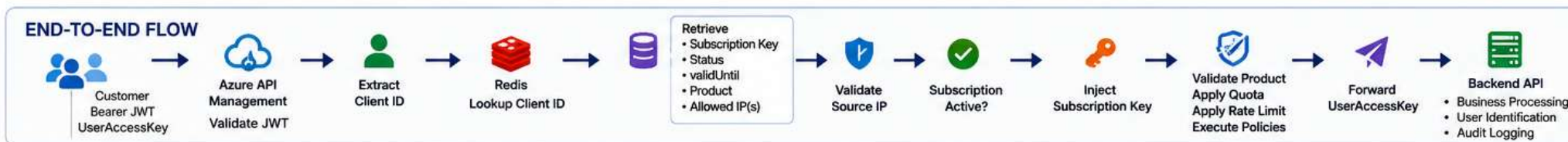
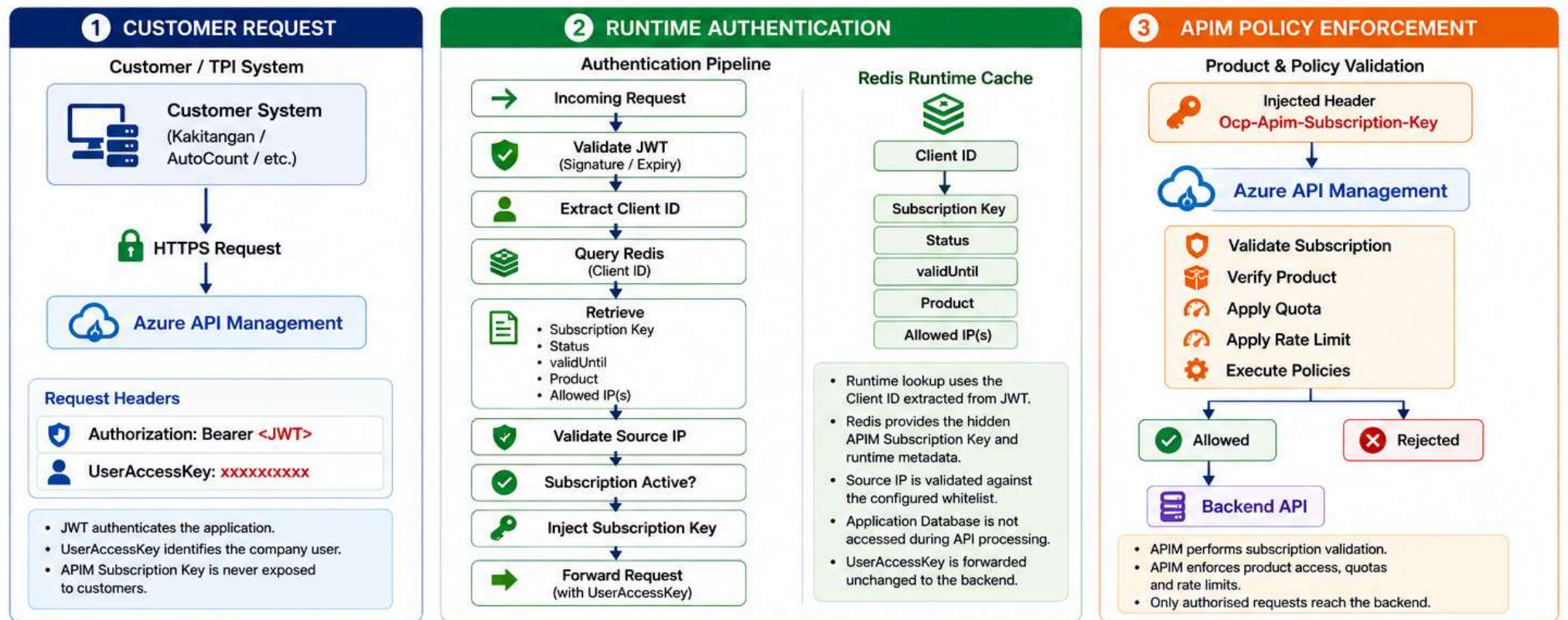


Credentials Assigned
to One User

Backend Credential Provisioning

Automated provisioning of API credentials, APIM subscription, and runtime mapping





COMPONENT	RESPONSIBILITY
Customer System	Sends JWT and UserAccessKey
Authentication Server	Issues OAuth Access Token (JWT)
Redis	Returns Subscription Key, Subscription Status, Validity, Product and Allowed IP(s) using Client ID
Azure API Management	Validates JWT, validates source IP, retrieves Subscription Key, validates subscription, enforces product policies, quota and rate limits
Backend API	Receives UserAccessKey and executes business logic
Application Database	System of record used during provisioning and Redis synchronisation



KEY TAKEAWAYS

- ✓ Customers send JWT and UserAccessKey only.
- ✓ APIM hides and injects the Subscription Key.
- ✓ Redis provides high-performance runtime validation.
- ✓ APIM enforces subscription, product, quota and rate limits.
- ✓ UserAccessKey is used by backend for business and audit.
- ✓ Application Database is not accessed during runtime.

Product Subscription Activation

Initial Product Selection, Approval and Subscription Activation

1 Available API Products

Product Catalogue



FREE

500 Calls / Month

RM 0



BASIC

10,000 Calls / Month

Monthly Subscription



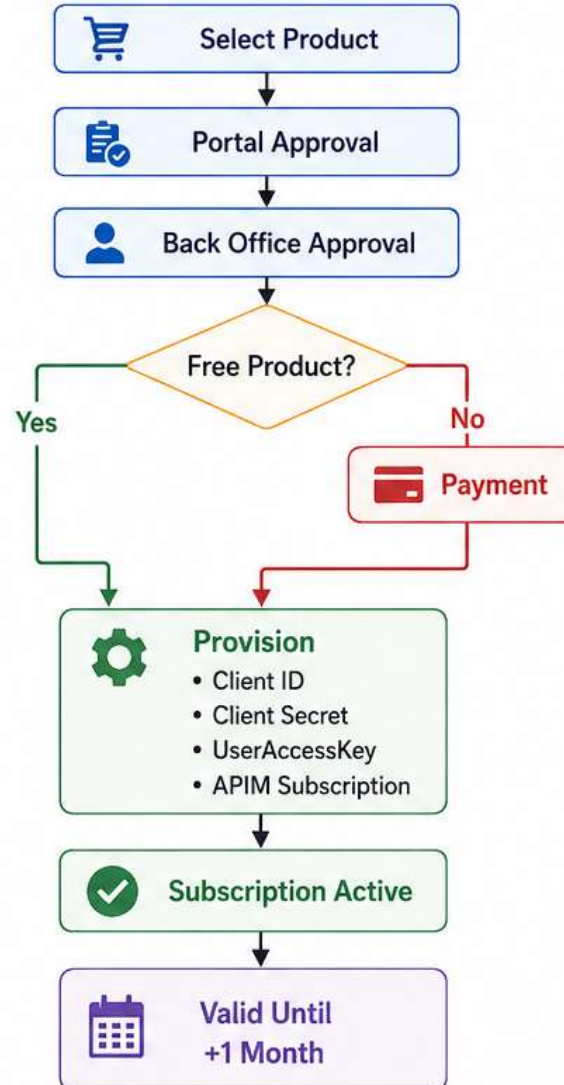
PREMIUM

100,000 Calls / Month

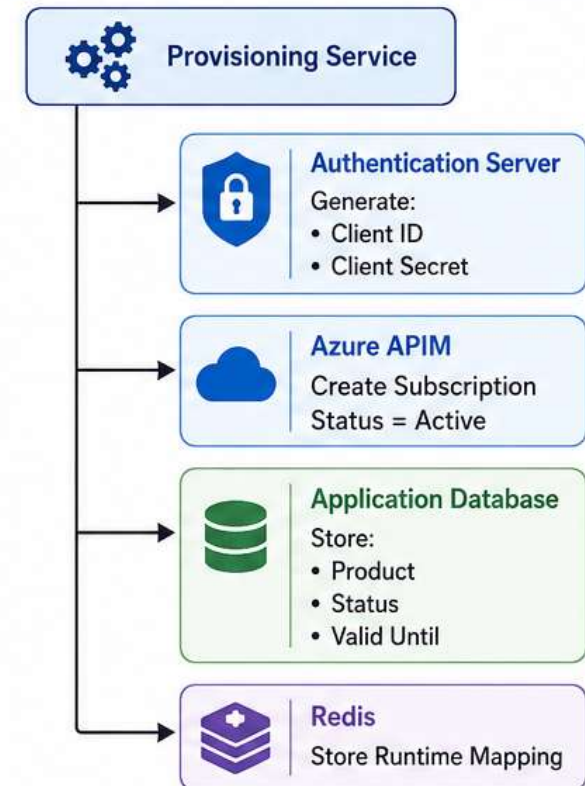
Monthly Subscription

- Customers select one Product during credential creation.
- Each Product provides different API quotas and policies.
- Product subscription is valid for one month from activation.

2 Product Subscription Workflow



3 Backend Activation (Provisioning Result)



- Customer credentials remain unchanged throughout the subscription period.
- Application Database is the system of record.
- Redis provides fast runtime lookup for API requests.
- Azure APIM immediately applies the Product's quota, rate limits and access policies.

EXPIRY REMINDER

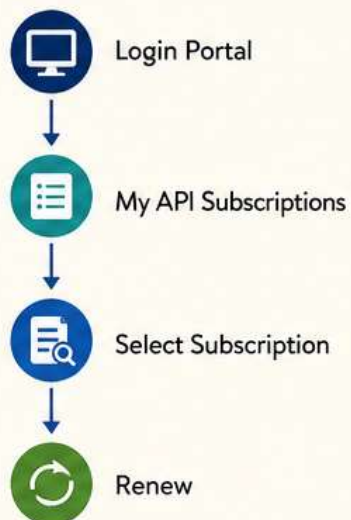
Automatic Reminder



- Reminder sent several days before expiry.
- Reminder schedule is configurable.
- Encourages renewal before service interruption.

CORPORATE ADMINISTRATOR

Renewal Initiation



- Current subscription information displayed.
- New validity period is calculated automatically.
- Customer reviews before submission.

Renewal Screen



RENEWAL SUBMITTED



- Renewal request created.
- Awaiting CFO approval.
- No Back Office approval required.

RENEWAL PROCESS OVERVIEW

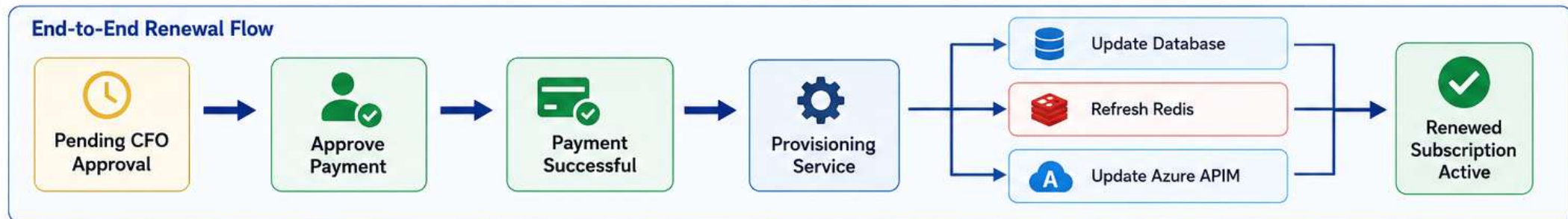
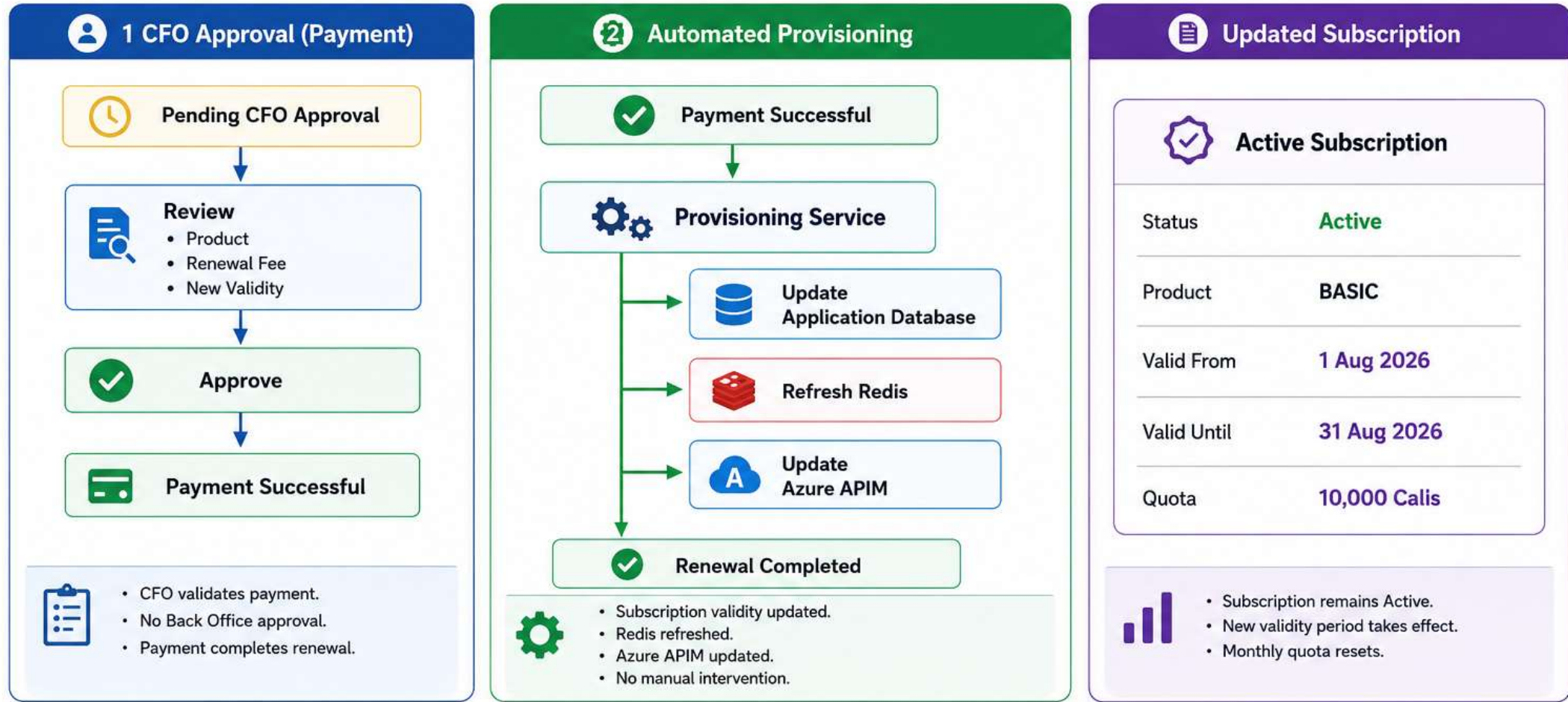


SPEAKER NOTES

- Reminder emails are automatically sent before subscription expiry.
- Both the Corporate Administrator and API User receive the notification.
- Only the Corporate Administrator can renew the subscription.
- The Portal displays the newly calculated renewal period before submission.
- After submission, the request proceeds directly to CFO approval.

API Subscription Renewal & Re-Provisioning

CFO Payment Approval and Automated Subscription Extension



API Subscription Expiry Batch Job

Automatic Synchronization of Expired API Subscriptions



SUBSCRIPTION EXPIRY SCHEDULER



Every day at 00:00



Subscription Expiry
Batch Job

- Runs on configurable schedule
- Queries Application Database
- Finds subscriptions where:
 - validUntil < Current Time
 - status = Active



BATCH PROCESSING FLOW



Application Database (CMS)



Find Expired Subscriptions
(validUntil < Now)



For Each Subscription



Update Redis
Status = Expired



Call Azure APIM
Update Subscription
Status = Expired



Update App Database
Status = Expired
Expired Time

REDIS UPDATE

- clientId
- subscriptionKey
- status = Expired
- validUntil



AFTER BATCH EXECUTION



Application DB
✓ Status = Expired



Redis
✓ Status = Expired



Azure APIM
✓ Subscription = Expired



Incoming API Request



Rejected

FLOW SUMMARY



1 Scheduler
Trigger



2 Query
App DB



3 Expired?



4 Yes



5 Update
Redis



Update
APIM Subscription
(Status = Expired)



7 Update
App DB Status



8 API Access
Blocked



SPEAKER NOTES

- The Application Database owns the subscription validity (validUntil).
- The batch job periodically identifies subscriptions that have passed their expiry date.
- Redis is updated first so authentication immediately recognizes the subscription as expired.

- Azure API Management subscription status is updated to Expired, ensuring quota and product access are disabled.
- Finally, the Application Database records the new subscription status and expiry timestamp.
- This keeps the Application Database, Redis, and Azure APIM synchronized without manual intervention.